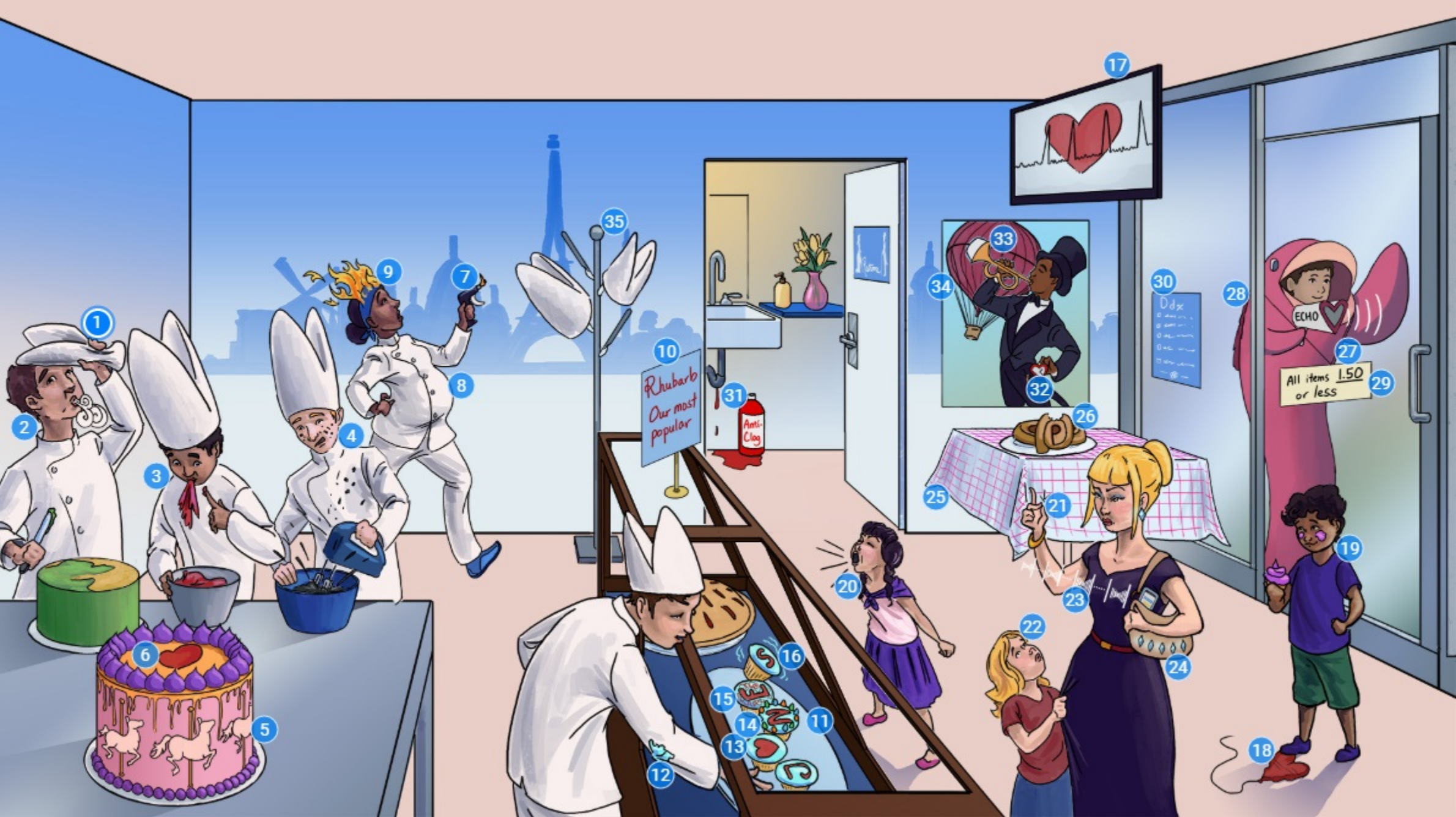




3.3 - Mitral Stenosis SOAP

1) Squeezed mitre hat:

Mitral Stenosis = narrowing of the mitral valve causing obstruction to blood flow from LA to LV

2) Baker blowing puff of air:

Dyspnea on exertion = most common presenting symptom of **MS** (**Chief Complaint**)

3) Baker coughing red frosting:

MS can present with hemoptysis due to increased pulmonary pressures (**Chief Complaint**)

4) Black frosting on face/body:

Embolic event (e.g., stroke) can be the first presentation of **MS** (**Chief Complaint**)

5) Horses on cake:

Ortner's syndrome = **voice hoarseness** in **MS** due to impingement of the recurrent laryngeal nerve by a significantly enlarged left atrium (**Chief Complaint**)

6) Abnormally large left heart:

Ortner's syndrome = voice hoarseness in **MS** due to impingement of the recurrent laryngeal nerve by a **significantly enlarged left atrium** (**Chief Complaint**)

7) Chef's torch with huge trigger:

MS symptoms can be **triggered** by any situation that increases CO or HR (e.g., exercise, fever, pregnancy, arrhythmias) (**HPI**)

8) Pregnant woman:

MS symptoms can be triggered by any situation that increases CO or HR (e.g., exercise, fever, **pregnancy**) (**HPI**)

9) Flame headband:

MS symptoms can be triggered by any situation that increases CO or HR (e.g., exercise, **fever**, pregnancy) (**HPI**)

10) Rhubarb - our most popular!:

Rheumatic heart disease = most common cause of **MS** (**PMHx**)

11) J♥NES cupcakes:

Major manifestations of **ARF** = JONES → **J**oints (arthritis), ♥ (carditis), **N**odules (subcutaneous), **E**rythema marginatum, **S**ydenham chorea (**PMHx**)

12) “J” cupcake with frosting on elbow:

“Joints” = migratory polyarthritis predominantly involving large joints (e.g., knees, ankles, elbows) (**PMHx**)

13) “♥” heart cupcake:

“Heart” = pancarditis that can involve the pericardium, myocardium, and endocardium (**PMHx**)

14) “N” cupcake with nodular candies:

“Nodules” = SUBCUTANEOUS nodules most commonly found on elbows (**PMHx**)

15) “E” cupcake with red lines on margins:

“Erythema marginatum” = pink, non-pruritic rash with red margins appearing on the trunk & limbs, sparing the face (**PMHx**)

16) “S” cupcake shaking & falling:

“Sydenham chorea” = neurologic disorder consisting of abrupt, nonrhythmic, involuntary movements & emotional changes (**PMHx**)

17) Irregularly irregular signal:

Left atrial enlargement caused by MS can lead to **AFib** → look for an irregularly irregular pulse on exam (**Vitals**)

18) Floppy heart balloon:

Severe **MS** can eventually lead to **right-sided heart failure** → look for elevated JVP, pitting edema, & hepatjugular reflux (**Physical Exam**)

19) Pink-purple patches on face:

Mitral facies = pink-purple patches on cheeks caused by cutaneous vasoconstriction from severe **MS** & decreased cardiac output (**Physical Exam**)

20) Extra loud first sister:

S1 is typically extra loud in **MS** (**Cardiac Auscultation**)

21) Snapping fingers:

Opening **snap** = high-pitched sound in early diastole (immediately after S2) best heard at the apex (**Cardiac Auscultation**)

22) Impatient second sister:

A shorter interval between S2 and opening snap indicates more severe **MS** (**Cardiac Auscultation**)

23) Murmur lines:

Classic **MS** murmur = Low-pitched rumbling mid-diastolic murmur best heard at the apex (**Cardiac Auscultation**)

24) Diastolic diamonds:

Classic **MS** murmur = low-pitched rumbling mid-**diastolic** murmur best heard at the apex (**Cardiac Auscultation**)

25) ECG tablecloth:

Common **ECG** findings in **MS** = AFib & LA enlargement (**ECG**)

26) P-shaped cinnamon roll with bite mark:

Left atrial enlargement caused by **MS** results in a broad, **notched P wave** in lead II known as “P mitrale” (**ECG**)

27) ECHO megaphone:

TTE = imaging modality of choice for diagnosis of **MS** → look for thickened, calcified mitral valve leaflets & “fish-mouth” valve orifice (**Echo**)

28) Fish costume:

TTE = imaging modality of choice for diagnosis of **MS** → look for thickened, calcified mitral valve leaflets & “**fish-mouth**” **valve orifice** (**Echo**)

29) “All items 1.50 or less”:

Severe **MS** is defined as a mitral area $\leq 1.5 \text{ cm}^2$ (**Echo**)

30) DDx clipboard:

For further workup of **Mitral Stenosis**, see Chronic Dyspnea DDx

31) Anti-Clog solution:

All patients with **MS** and risk factors for thromboembolism (i.e., AFib, prior embolic event, left atrial thrombus) should receive **anticoagulation** with warfarin or heparin **(1)**

32) Heart-shaped watch:

Heart rate control with beta blockers can be helpful in patients with **MS** and symptoms due to tachycardia (e.g., AF with RVR, exercise) **(2)**

33) Muted beta-trumpet:

Heart rate control with **beta blockers** can be helpful in patients with **MS** and symptoms due to tachycardia (e.g., AF with RVR, exercise) **(2)**

34) Hot air balloon:

Percutaneous mitral balloon commissurotomy is the initial intervention of choice in most patients with severe **MS** & symptoms **(3)**

35) Mitre hat stand with scalpel-shaped pegs:

Mitral valve surgery (repair / replacement) is indicated in patients who are not candidates for or have previously failed PMBC **(4)**